

Equation wrangling: How I learned to stop worrying about large equational sets and love automated deduction

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Quasigroups

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A *quasigroup* is a set with three binary operations, $\cdot$, $/$, $\backslash$, satisfying:

$$x \cdot (x \backslash y) = y = (y / x) \cdot x$$

$$x \backslash (x \cdot y) = y = (y \cdot x) / x$$

Loops

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2. *Moufang loops*: $x \cdot (y \cdot (x \cdot z)) = ((x \cdot y) \cdot x) \cdot z$
2. *(left) Bol loops*: $x \cdot (y \cdot (x \cdot z)) = (x \cdot (y \cdot x)) \cdot z$

Notation simplifier

Juxtaposition takes priority

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Left Bol: $z(y \cdot zx) = (z \cdot yz)x$

Right Bol: $x(zy \cdot z) = (xz \cdot y)z$

Moufang: $z(x \cdot zy) = (zx \cdot z)y$

$$(xz \cdot y)z = x(z \cdot yz)$$

$$(z \cdot xy)z = zx \cdot yz$$

$$z(xy \cdot z) = zx \cdot yz$$

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Bol-Moufang identity

60 such identities; 14 distinct loop varieties.

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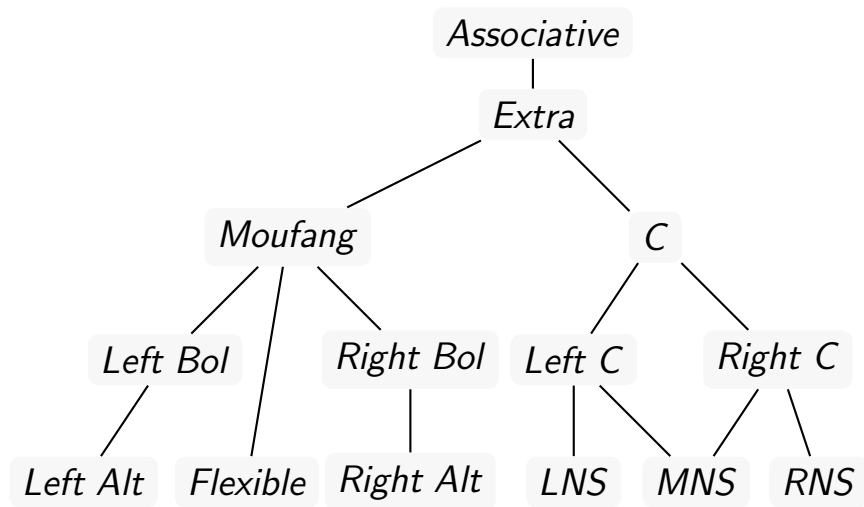


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1215 such identities; 48 varieties of loops.

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1. Automated deduction and finite model building
2. Hasse diagram too complicated for slide
3. Stamp collecting, sure; but structure emerges

Frute loops

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Theorem (Drapal and P., '19)

Frute loops are precisely the products $O \times H$, where O is a commutative Moufang loop in which all elements are of odd order, and H is a 2-group such that the derived subloop H' is of exponent two and $H' \leq Z(H)$.

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Theorem (P. and Shcherbacov, '10)

Cheban loops are centrally nilpotent of class at most two.

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Theorem

Let L be a left C loop, i.e., a loop in which $x(y \cdot yz) = (x \cdot yy)z$. Then, $N_{\lambda}(L) = N_{\mu}(L)$ is normal in L .

Mate loops

left Mate: $(x \cdot xy)z = (yz \cdot x)x$

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Theorem

Let Q be a left mate loop. Then Q is a left C loop; moreover, $Q/N_\lambda(L)$ is commutative.

Thanks!

Thank you for your kind attention.